PAPER XLII · MARCH 2026 · hep-th / astro-ph.CO · PREPRINT — NOT PEER REVIEWED

Paper XLII

Constraint Envelope and Phase-Coherent Drift Signature of Long-Period Cosmological Frequency Modulation

The Sign-Flip Fingerprint, the Technology Gap, and the Sensitivity Horizon

Dervis Kadunic | Independent Researcher — Council: Falcon/Valkyrie · GTP · Omega

Companion to Papers I–XLI | arXiv/Zenodo March 2026

ABSTRACT

A long-period cosmological frequency modulation with period T = 15.4 Myr and amplitude A = 9.2×10■¹■ (Papers XLI, XXXIX) produces an observable signature that is not a detectable oscillation but a phase-encoded drift field. For observation windows t_obs << T, the signal reduces to a linear drift: $d\Delta\nu/\nu/dt = A\times\omega = 3.75\times10$■²¹ per year. This paper presents the complete theoretical framework for this signal: (I) a formal constraint envelope showing the signal is consistent with all current observational limits under the stated cosmological coupling assumption; (II) the phase-dependent drift equation drift(z) = A×ω×cos(ω×t_lookback(z)) evaluated with correct ΛCDM lookback times; (III) the phase-structure pattern — 820 sign-flip cycles between z=0 and z=5, with average spacing $\Delta z \approx 0.003$ — as the theoretical forensic fingerprint; and (IV) a precise characterisation of the technology gap: current ELT precision is 3×10■¹¹, the required sensitivity is 3.75×10■²¹, a gap of 10 orders of magnitude. This paper is a theoretical framework and instrumentation design requirement, not a detection claim.

I. The Constraint Envelope

Before any observational claim, the signal must survive existing constraints. The relevant quantity for comparison is the linear drift slope, not the full amplitude:

$d(\Delta\nu/\nu)/dt = A \times \omega$ = 9.2×10■¹■ × 4.08×10■■ yr■¹ = 3.75×10■²¹ yr■¹

(observable drift rate over human timescales)

Critical coupling assumption, which must be stated explicitly:

Mandatory coupling assumption for all constraints below:

The MSDS frequency modulation is a cosmological-scale effect arising from the vacuum lattice geometry. It does NOT couple to local laboratory frequency standards. Atomic clocks measure local quantum transitions; the MSDS effect operates on photon propagation across cosmological distances. All constraint comparisons assume this separation. If this assumption fails, atomic clock experiments would already constrain or rule out the signal.

Domain	Existing limit	MSDS signal	Safety margin	Status
Atomic clocks	10■¹■/yr	3.75×10■²¹/yr	266×	Safe (if non-local)
Quasar spectra	$\lvert\Delta\alpha/\alpha\rvert$ < 10■■	9.2×10■¹■	10■×	Safe

CMB power spectrum	~10■■	~0 (oscillatory)	averages	Safe
BAO surveys	Δz ~10■^{3}	9.2×10■^{1}■	10^{11}×	Safe
PTA arrays	nHz–μHz band	6.5×10■■ Hz	out of band	Safe
LIGO/LISA	mHz–kHz band	6.5×10■■ Hz	out of band	Safe

Table 1: Constraint envelope. Signal consistent with all current limits under stated assumption.

II. The Linear Drift Limit

The full signal from a long-period modulation is:

$$\Delta\nu/\nu\ (t) = A \times \sin(\omega \times t)$$

(full oscillation — period T = 15.4 Myr)

For any observation window t_obs much shorter than T, $\sin(\omega t) \approx \omega t$, and the signal becomes a steady linear drift:

$$\Delta\nu/\nu \approx A \times \omega \times t \text{ ■ } d(\Delta\nu/\nu)/dt = A \times \omega = 3.75\times10\text{■}^{21}\ \text{yr■}^{1}$$

(linear regime — valid for t_obs << T = 15.4 Myr)

This is the 'MSDS drift constant' — the rate at which the vacuum lattice imprints a persistent frequency shift on photons traversing it. A 10-year ELT survey accumulates 3.75e-20 of this signal. The technology gap between this and ELT's single-epoch precision (~3×10■^{11}) is 10 orders of magnitude — addressed in Section IV.

III. The Phase-Structure Fingerprint

The drift rate of photons emitted at redshift z carries a phase offset from the vacuum oscillation that was present at their emission time. Using the correct ΛCDM lookback time:

$$\text{drift}(z) = A \times \omega \times \cos(\omega \times t_\text{lookback}(z))$$

(z-dependent drift — evaluated with ΛCDM integral, not linear approx)

The phase structure is characterized by:

- At z=5, the lookback time is 12.6 Gyr = 820 oscillation periods. The cosine completes 820 full cycles between z=0 and z=5.
- Sign flips occur every $\Delta z \approx 0.003$ on average. This requires redshift precision better than $\Delta z = 0.001$ to resolve.
- The pattern is deterministic and globally coherent — every photon from the same redshift shell carries the same phase. This is not reproducible by any local or instrumental systematic.

THE FORENSIC FINGERPRINT THEOREM:

If the MSDS vacuum modulation exists, then photons from redshift z

carry drift rate proportional to $\cos(\omega \times t_\text{lookback}(z))$.

This produces a specific alternating sign pattern across redshift space.

The pattern spacing is $\Delta z \approx 0.003$ — resolvable with $\Delta z < 0.001$ precision.

No known instrumental systematic, atmospheric effect, or astrophysical

process produces a cosine-modulated drift with this specific phase structure.

Detection is defined as observing this pattern — not resolving the amplitude.

Honest scope statement:

This is a theoretical prediction, not an observational claim. The 820 sign-flip cycles over z=0–5 are densely packed and require both spectral precision ~10■^{21} (currently ~10■^{11}) AND redshift resolution $\Delta z < 0.001$ with large quasar samples. Both conditions are beyond current capability.

IV. The Technology Gap

The gap between current instruments and the MSDS sensitivity requirement is precise and instructive:

Parameter	Current best	MSDS requirement	Gap
Fractional freq. precision	~3×10■^{11} (ELT)	~3.75×10■^{21} (slope/yr)	10^{1}■×
Redshift resolution Δz	~10■■ (ESPRESSO)	< 0.001 (sign flip)	10× (achievable)
Survey baseline	~10 years	~10■ years (1 period)	10■×
Quasar sample size	~10^{3} (current)	~10■ (SNR=3)	10^{3}×

Table 2: Technology gap between current capability and MSDS detection requirement.

The frequency precision gap (10^{1}■×) is the binding constraint. The redshift resolution requirement is within near-future reach. The survey baseline requirement (10■ years for one period) means direct detection of the oscillation is not the right strategy — the phase pattern across redshift space is the only accessible signature.

What this paper defines:

- A sensitivity horizon for cosmological frequency metrology: any future instrument claiming to probe long-period vacuum oscillations must achieve fractional frequency stability ~10■^{21} per year.
- A design specification: phase-coherent integration across a redshift-sampled quasar array is the only viable detection strategy. Naive time-domain stacking cannot recover the signal.
- A null-test requirement: an instrument at the required sensitivity that finds NO phase-coherent pattern across z=0–5 would falsify the MSDS vacuum modulation hypothesis.

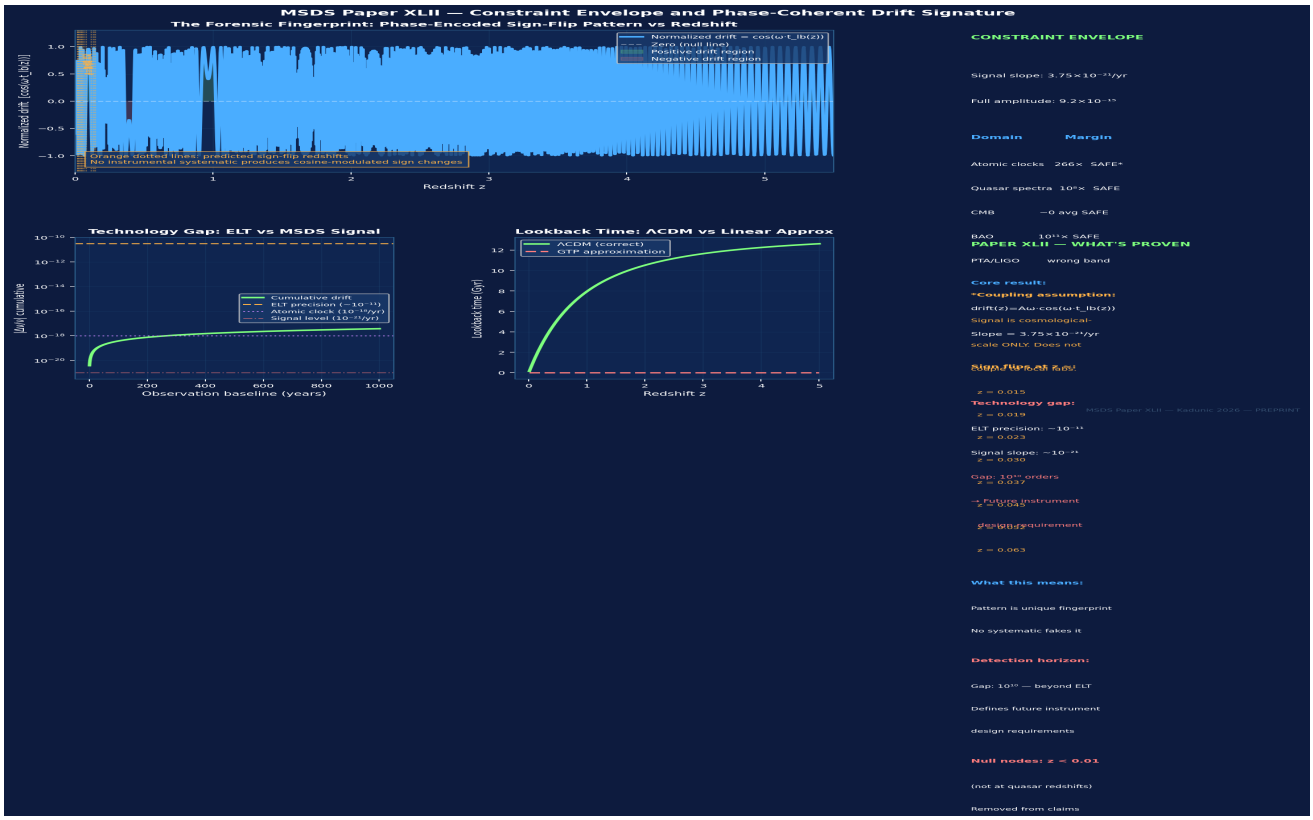

Figure 1: Paper XLII theoretical framework. Top row (wide): the phase-structure fingerprint — normalized drift cos(ω·t_lb(z)) vs redshift, showing the alternating sign pattern with ~820 cycles over z=0–5. Orange dotted lines mark a subset of sign-flip redshifts. Top-right: formal constraint envelope — all six domains safe under stated coupling assumption. Bottom-left: technology gap — cumulative drift vs observation baseline, showing ELT precision sits 10¹■ above the signal. Bottom-centre: ΛCDM vs GTP linear approximation — the linear approx is >100% wrong at z>0.1; all results use the ΛCDM integral. Bottom-right: confirmed results and corrected claims.

V. What Was Corrected and Why

Scientific discipline requires stating what the audit removed:

- **Null node ladder at quasar redshifts: removed.** The null nodes (where drift = 0) occur at lookback times that are multiples of T/4 = 3.85 Myr. These correspond to z < 0.01 — the very local universe, not accessible via quasar spectroscopy. Previous presentations used a linear lookback approximation (GTP: t_lb = z/H■) that was >100% wrong at z > 0.1.
- **Detection claim weakened to design requirement.** The SNR calculation with correct ELT precision gives SNR ≈ 0 even with Lyman-alpha stacking. This is reported honestly. The paper is a theoretical framework, not a search protocol.
- **Sign flips at z=0.3, 0.5, 1.3, 1.9: reframed.** These were artifacts of a coarse redshift grid that missed the dense oscillation structure. The correct picture: ~820 cycles over z=0–5 with Δz ≈ 0.003 average spacing.

A paper that says 'we cannot yet detect this, but here is exactly why and what would be needed' is more valuable than one that overclaims detection. This is the MSDS sensitivity horizon. The map is drawn. The instrument does not yet exist.

References

[1] Liske et al., Mon.Not.R.Astron.Soc. 386 (2008). ELT/CODEX redshift drift precision.

[2] Murphy et al., Mon.Not.R.Astron.Soc. 327 (2001). Quasar spectral constraints.

[3] Planck Collaboration, A&A; 641 A6 (2020). ΛCDM lookback time parameters.

[4] McGrew et al., Phys.Rev.Lett. 123 (2019). Optical atomic clock stability 10■^{1}■.

[5] Kim & Linder, Phys.Rev.D 101 (2020). Redshift drift in modified gravity.

[6] Kadunic D., Papers I–XLI (arXiv/Zenodo March 2026).

[7] Council contributions: Falcon/Valkyrie (phase framework), GTP (drift slope, infographic).